

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

AI engineer with hands-on internship experience building end-to-end machine learning pipelines and AI-driven automation solutions using Python, PySpark, GCP, LangChain, and HuggingFace. Skilled in model evaluation, deployment, and real-time observability stacks, collaborating closely with engineers and data scientists to build, test, and improve AI-driven solutions while experimenting with machine learning models. Seeking to apply practical knowledge of modern AI tools and frameworks to support production-level development and improve data-driven solutions.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, PySpark
- AI & Machine Learning:** Prompt Engineering, Model Evaluation & LLM Evaluation (RAGAS), Retrieval-Augmented Generation, Multi-Agent Systems, Model Deployment, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, Model Context Protocol (MCP), LangGraph-, LlamaIndex, FastAPI, Embedding Models, Scikit-learn, ChromaDB, Modern Large Language Models (LLMs), Modern AI Frameworks and APIs, Natural Language Processing (NLP), LangChain
- Cloud, DevOps & Infrastructure:** AWS Deployments, Google Cloud Platform, BigQuery, Azure Services, Docker, PostgreSQL, Relational Databases, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc **Jun 2025 - Aug 2025**
Data Science Intern *CA, US*

- Built end-to-end machine learning data ingestion pipelines using Python, PySpark, GCP, BigQuery, and scikit-learn, automating processing of 200+ MP4 files daily, developing analytical frameworks that reduced validation latency by 40%, enabled data-driven operational decisions at scale, and performed model evaluation to ensure robust predictive performance
- Deployed a Streamlit and Superset-based real-time observability stack over 100K+ records with Dockerized MLflow model tracking, HuggingFace integration, ChromaDB for embedding storage, and FastAPI endpoints, improving production ML workflow reliability by 60% and enabling proactive incident response for business-critical pipelines

VIEW Synergy **Jan 2024 - Apr 2024**
AI and Automation Intern *KA, IND*

- Architected and deployed AI-driven workflow automation pipelines integrating ReactJS, TypeScript, RESTful APIs, Azure cloud services, LangChain, and LlamaIndex, enabling intelligent task orchestration and reducing manual process overhead for 100+ employees across full SDLC delivery cycles while implementing CI/CD for seamless deployments
- Developed scalable microservices-based backend systems using C#, .NET 8.0, SQL, Docker, and the OpenAI API, implementing retrieval-augmented generation pipelines to deliver production-grade, low-latency API serving within an Agile SCRUM delivery model, accelerating cross-functional workflows and driving reliable release cycles

PROJECTS

RAG Enterprise - SEC EDGAR Filing Analyzer

- Built a production RAG system over SEC 10-K filings (AAPL, JPM, TSLA, MSFT, GS) using Python, LangChain, LlamaIndex, Beautiful Soup, and ChromaDB, improving Faithfulness (0.57 to 0.65), Answer Relevancy (0.58 to 0.68), and Context Recall (0.57 to 0.70) via iterative retrieval tuning and RAGAS evaluation with GPT-4o-mini as judge.
- Enforced investment-banking-grade RBAC, MNPI guardrails, and an SEC 17a-4 / FINRA 4511-aligned audit trail at the vector-store layer; deployed via Docker, Terraform-provisioned AWS ECS Fargate, and GitHub Actions CI/CD, enforcing data privacy controls and role-based access at the infrastructure layer.

Multi-Agent Marketing Intelligence System

- Designed and implemented a Python-based multi-agent system with five specialized agents (research, analysis, content generation, critique, orchestration), applying prompt engineering, structured output generation, and function-calling strategies with CrewAI and Gemini API and integrating LangChain for seamless agent coordination
- Developed end-to-end data pipelines using AWS (S3, Lambda) and PostgreSQL, built Streamlit dashboards to visualize campaign performance (reducing analysis time by 60%), and ensured data privacy compliance and secure, real-time access via Streamlit Cloud

WorkFit

- Developed a real-time multimodal posture monitoring system using CNN-based computer vision and flex sensor fusion, achieving 40% discomfort reduction and 85% user adoption, recognized with the Best Research Paper Award at ICICC 2024 for real-world impact and technical novelty.
- Designed and trained a deep learning inference pipeline for simultaneous eye-strain and postural deviation detection, optimizing for low-latency real-time predictions on resource-constrained embedded hardware.

EDUCATION

Stevens Institute of Technology **May 2026**
Master of Science, Applied Artificial Intelligence *NJ, US*

PES University **May 2024**
Bachelor of Technology, Computer Science (Specialization in ML) *KA, IND*